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Integrated Biomedical and Clinical Sciences OMFS 513

Immunodeficiency

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Learning Objectives

1. Define the primary (congenital) immunodeficiency.
2. Explain the etiology and clinical manifestations of Chronic granulomatous disease (CGD) and Leukocyte adherence deficiency (LAD).
3. Explain the etiology and clinical manifestations of different clinical cases of related to severe combined immunodeficiencies (SCID).
4. Highlight some therapeutic approaches for the congenital immunodeficiency.
5. Define the secondary (acquired) immunodeficiency.
6. Explain the immunopathogenesis of Acquired Immunodeficiency Disease (AIDS).
7. Discuss the mechanism of immunodeficiency caused by Human Immunodeficiency virus (HIV).
8. To know the drugs used to treat AIDS

What is Immunodeficiency ?

It is a condition in which the immune system fails to fight infectious diseases and cancer due to the absence or deficiency of the components of innate or adaptive immunity.

The importance of understanding immunodeficiency for dentists

1. **Risk Management:** Patients with immunodeficiency are at higher risk for infections → Routine dental procedures should be properly managed.
2. **Treatment modifications:** Treatment plans should be adapted to immunocompromised patients compared to healthy individuals.
3. **Infection control:** Dentists should implement more strict infection control measures by understanding how immunodeficiency affects the quality of life and oral health of their patients.
4. **Oral manifestations diagnosis:** Many immunodeficiency are associated with oral signs such as candidiasis therefore, early recognition of these signs can lead to timely medical referrals.

Overview

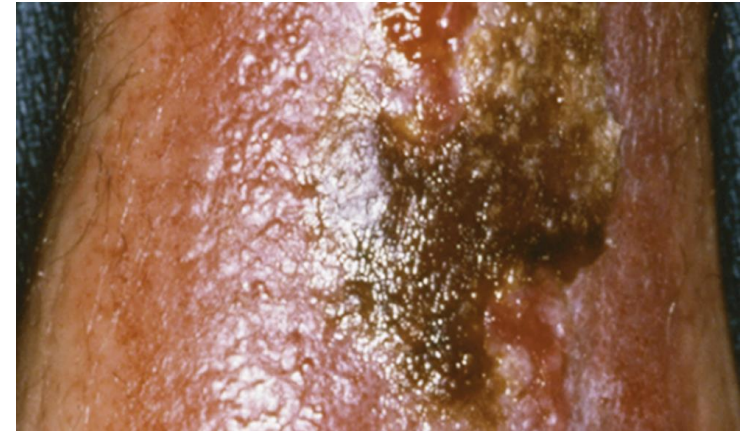
- ❑ Immunodeficiency may result from:
 - ✓ Defects in the **effector mechanisms of innate immunity**.
 - ✓ Defects in **lymphocyte development or activation**.
- ❑ The principal consequence of immunodeficiency is **increased susceptibility to infections and cancer**.

Example 1: **Opportunistic infection** with **Candida albicans**.

(The top picture shows fungal thrush due to infection with *Candida albicans* (aka, Candidiasis)).



Example 2: **Cancers** caused by **oncogenic viruses**, such as the Epstein- Barr virus (EBV) and human papilloma viruses (HPVs). (The bottom picture shows AIDS patient with Kaposi's Sarcoma).



❑ Immunodeficiency disorders are classified into:

- ✓ **Primary (congenital) Immunodeficiency.** Due to **inherited gene mutations** affecting the development or functionality of the immune cells.
- ✓ **Secondary (acquired) Immunodeficiency.** Deficiencies of the immune system often develop **because of abnormalities** that are not **genetic but acquired during Life.**

Primary (Congenital) Immunodeficiency

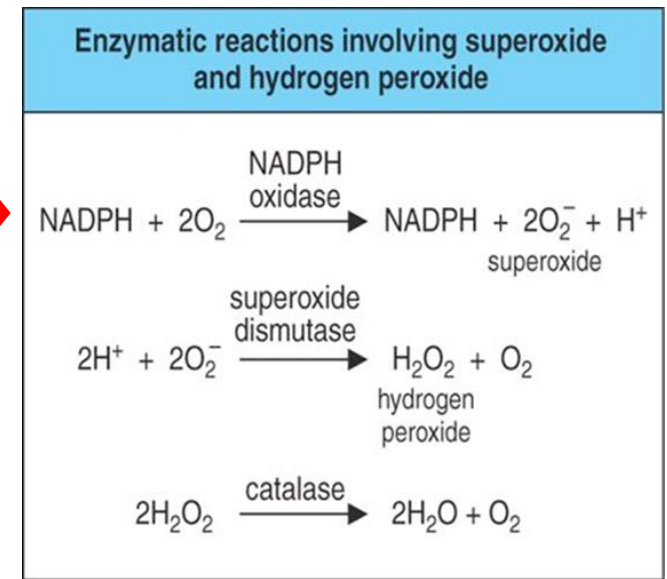
Defects in innate immunity

Deficiencies in phagocyte function

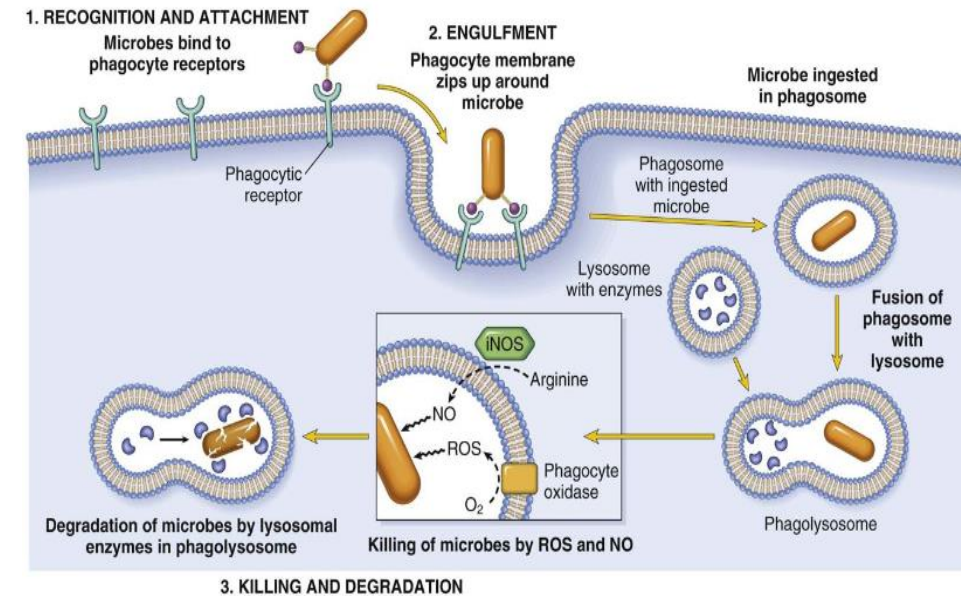
- ✓ Chronic granulomatous disease (CGD)
- ✓ Leukocyte adherence deficiencies (LADs)

1. Chronic granulomatous disease (CGD):

- ❑ It is a rare disease, estimated to affect approximately 1 in 200,000 individuals in the United States in which **two-thirds of cases** show an **X-linked recessive pattern of inheritance** and the remainder is autosomal recessive.
- ❑ **Genetic defect:** is caused by **mutations** in components of the **NADPH oxidase enzyme (Phagocyte Oxidase, aka, PHOX)** resulting in **defective production of superoxide anion (No respiratory burst)**, one of several reactive oxygen species (*ROS*) that constitute a major **microbicidal mechanism** of phagocytes.
- ❑ **PHOX** is a multi-subunit enzyme that is **assembled** in activated phagocytes, mainly in the **phagolysosome membrane**.
- ❑ **PHOX** is activated by many stimuli, including **IFN-γ** and signals from **TLRs**.



Source: Parham Fig. 3.17

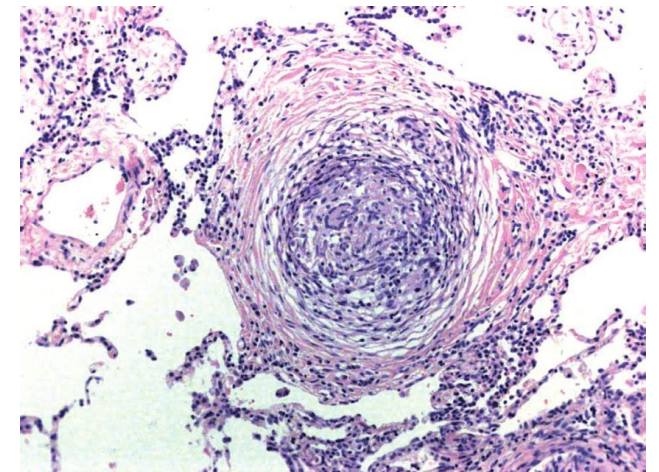
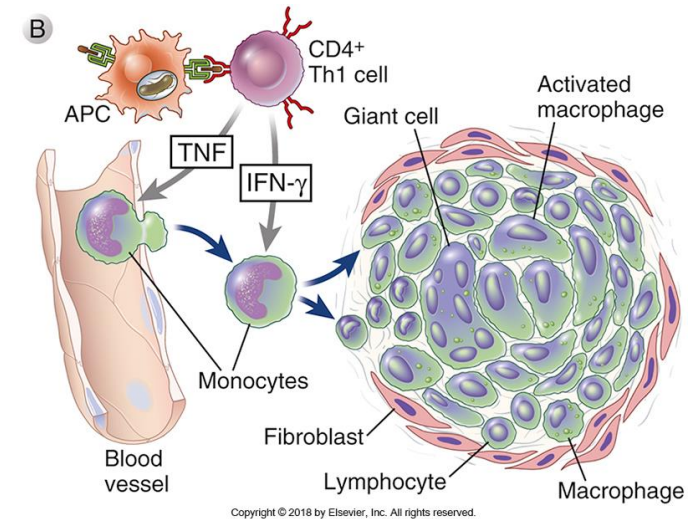


Source: Robbins Fig. 3.8

❑ **Immunopathogenesis:** Defective production of ROS → defective respiratory burst in neutrophils, macrophages and monocytes → a failure to kill phagocytosed microbes → infections are not controlled → stimulate chronic cell-mediated immune responses → Th1-mediated macrophage activation → formation of granulomas composed of activated macrophages, which try to eliminate the microbes.

❑ **Clinical manifestations:** CGD is characterized by recurrent infections with catalase +ve bacteria (*Staphylococcus aureus*) and fungi (*Aspergillus* and *Candida* Sp.), usually from early childhood leading to:

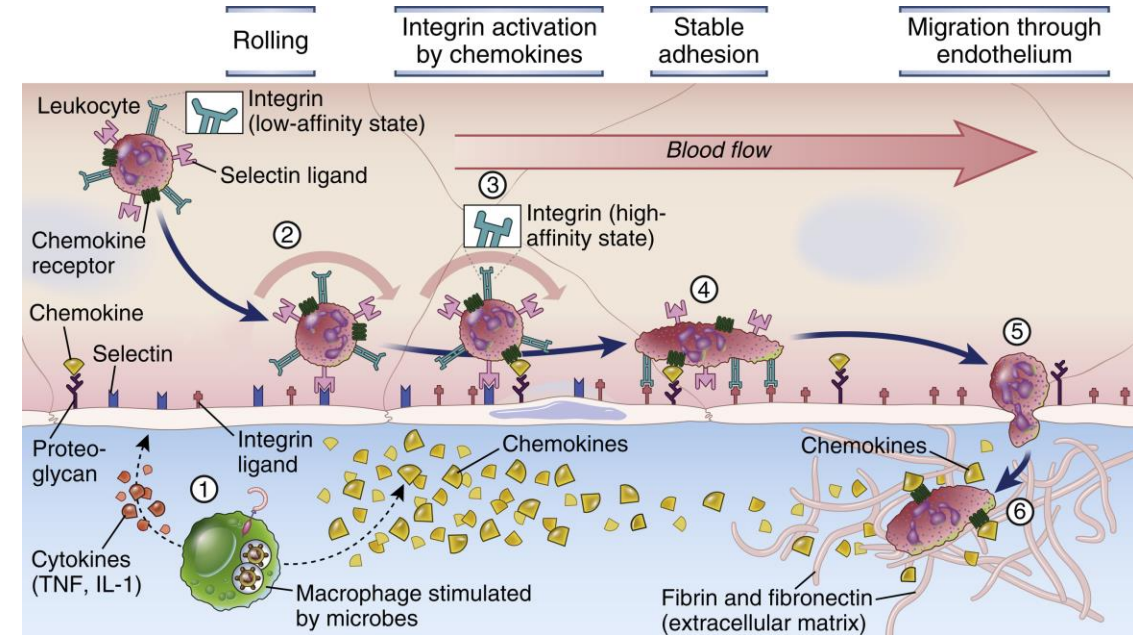
- ✓ Infections of skin (abscess), infections of respiratory tract (pneumonia).
- ✓ Formation of granulomatous lesions (e.g., lymphadenopathy if granulomas are formed in the lymph nodes).
- ✓ Bacteremia, fungemia.



Source: Cellular and Molecular Immunology, 9th edition. Ch. 19

2. Leukocyte adhesion deficiencies (LADs)

- ❑ The leukocyte adhesion deficiencies are a group of **autosomal recessive disorders**.
- ❑ These diseases are characterized by **defects in leukocyte and endothelial adhesion molecules** → a **failure of leukocytes**, particularly neutrophil, binding to blood vessel wall and **recruitment to sites of infection or injury** (**Failure of Leukocyte extravasation**).



Source: Cellular and Molecular Immunology. 9th edition. **Ch. 3**

- ❑ Patients with LADs suffer from **severe and recurrent** bacterial and fungal **infections** starting early in life in addition to **compromised wound healing**.
- ❑ Depending on the type of leukocyte adhesion molecule deficiency, **LADs are divided in 3 categories** (LAD-1, 2 and 3).

LAD-1

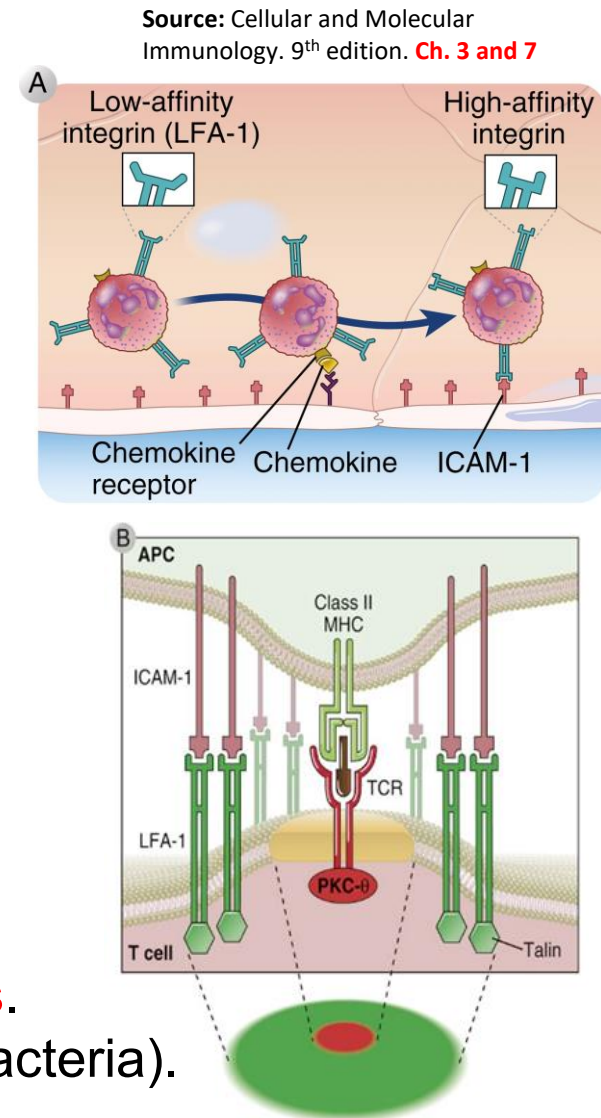
❑ **Genetic defect:** mutations in the CD18 gene leading to the **absence or reduced expression of the $\beta 2$ integrins** (heterodimers of CD18 and the CD11 family of glycoproteins).

❑ The $\beta 2$ integrins include leukocyte function–associated antigen 1 (**LFA-1**), **MAC-1** (CD11b) which participate in the **adhesion of leukocytes to endothelial cells**, and the binding of T lymphocytes to antigen-presenting cells (**Immune synapse**).

❑ The **defect leads to failure of leukocyte extravasation**.

❑ Clinical manifestations:

- ✓ Recurrent **bacterial and fungal infections of the skin and mucosal membranes**.
- ✓ **No pus formation** (pus is primarily composed of neutrophils, dead cells and bacteria).
- ✓ **Impaired wound healing**.
- ✓ **Delayed umbilical cord separation** (which normally occurs because of inflammation and neutrophil infiltration). (**used for diagnosis**)
- ✓ **Leukocytosis** is common (Leukocytes are trapped in the circulation and don't extravasate to tissues).



Defects in adaptive immunity

1. Severe combined immunodeficiencies (SCID)

- ✓ The DiGeorge Syndrome
- ✓ X-linked SCID
- ✓ SCID caused by defects in V(D)J recombination

2. B cell immunodeficiencies

- ✓ X-linked agammaglobulinemia
- ✓ Selective IgA deficiency

3. T cell immunodeficiencies

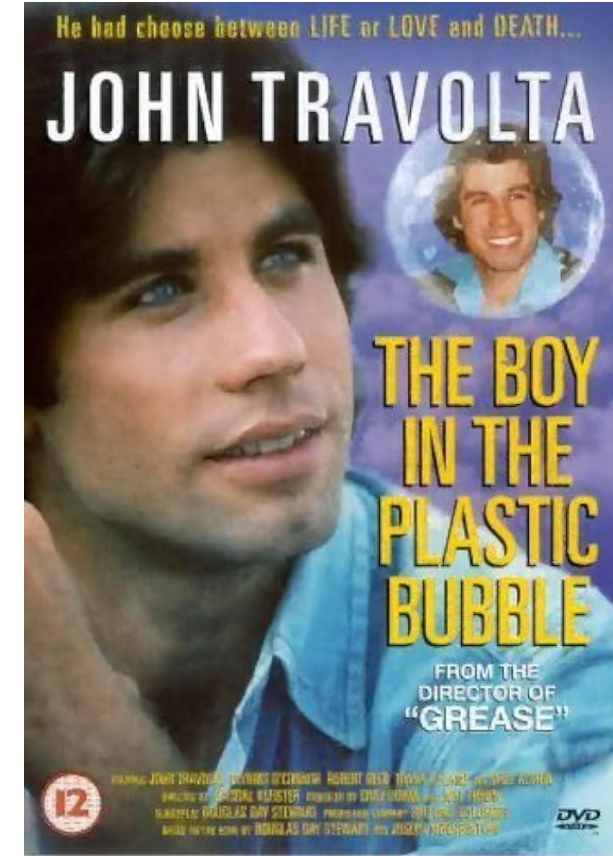
- ✓ The X-linked hyper-IgM (HIGM) syndrome

Defects in adaptive immunity

- ❑ Immunodeficiencies that **affect both humoral and cell-mediated immunity** are called **severe combined immunodeficiencies (SCID)**.
- ❑ The **process of T and B lymphocyte maturation** from hematopoietic stem cells to functionally competent mature lymphocytes involves **proliferation** of lymphocyte progenitors, **rearrangement of antigen receptor genes**, and **selection of cells with useful specificities**.
- ❑ Mutations in the **genes involved in these different steps of lymphocyte development** causes SCID.
- ❑ SCID may result from **impaired T lymphocyte development with or without defects in B cell maturation**. When there is no block in B cell development, the **defect in humoral immunity** is due to the **absence of T cell help** (secondary effect).

SCID

(The Baby bubble syndrome)



David Vetter
(The boy in the bubble)

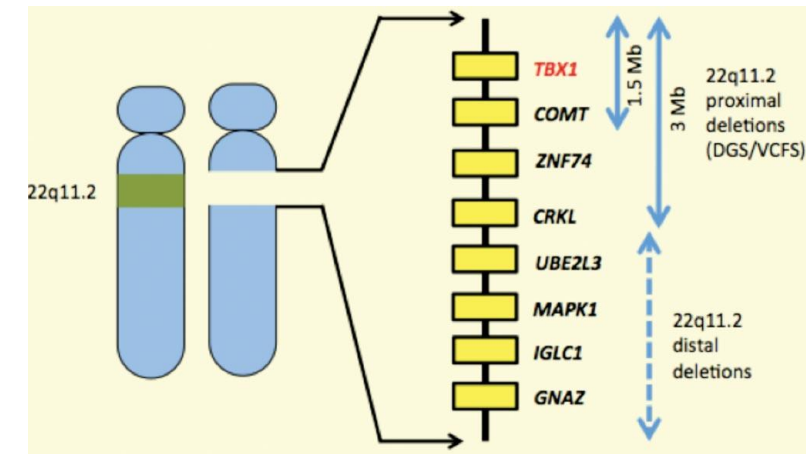
Clinical manifestations of SCIDs

- ❑ SCID patients suffer from severe infections that may be life-threatening, including pneumonia, meningitis, and bacteremia.
- ❑ Many viruses cause serious disease in patients with SCID.
 - ✓ Chickenpox (varicella) infection that is usually limited to the skin and mucous membranes in immunologically normal children, can progress to involve the lungs, liver, and brain in SCID patients.
 - ✓ Cytomegalovirus (CMV), which is present as a latent infection in most people, may be easily transmitted, often from mothers to their SCID offspring, and cause fatal pneumonia in newborns.
- ❑ Children with SCID often suffer from GIT infections (Rotavirus), recurrent respiratory infections, and a general failure to thrive.
- ❑ Children with SCID may also develop infections caused by live attenuated vaccines, such as varicella vaccine which are not harmful in children who have normal immunity. (live attenuated vaccines are contraindicated in SCID cases)
- ❑ Absence of tonsils.
- ❑ Children with SCID show thrush in mouth and diaper area (due to uncontrolled fungal infection)

Clinical cases of SCID

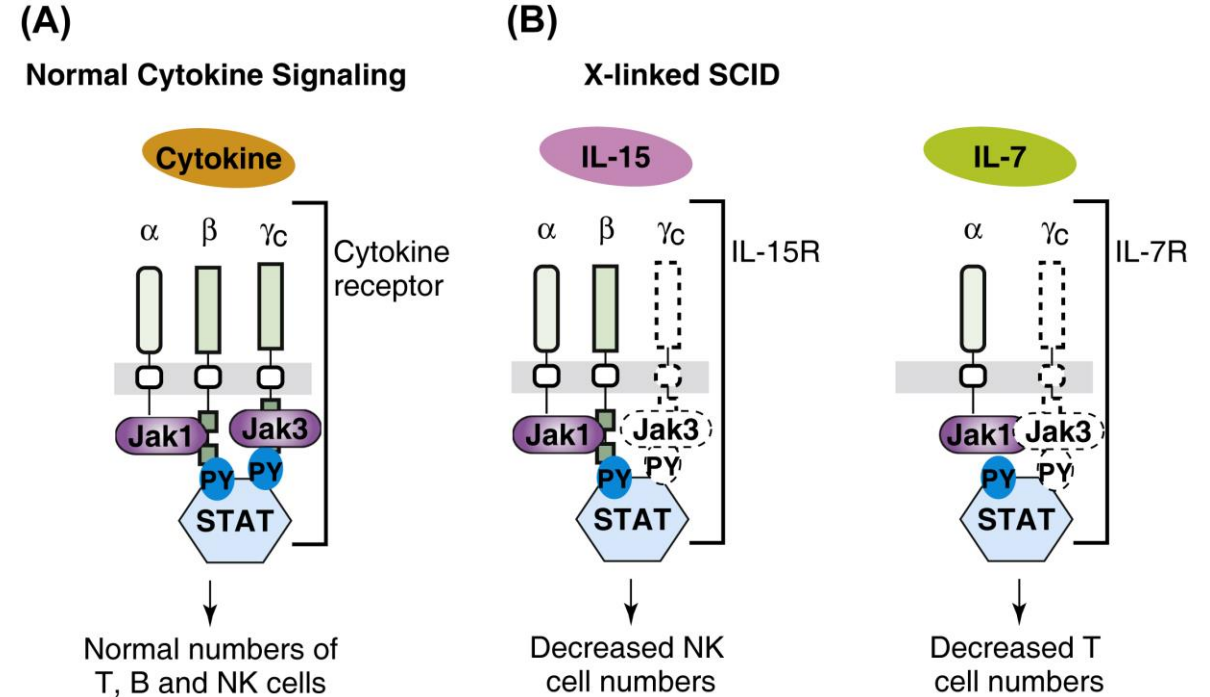
1.The DiGeorge Syndrome:

- ❑ is the form of SCID due to **defective thymic epithelial development** resulting in defective **development of the thymus**.
- ❑ **Genetic defect:** This disorder typically results from various **deletions** in a region on **chromosome 22q11** containing up to 50 genes, with the **T-box transcription factor (TBX1)** thought to be most influential. **TBX1** required for normal **thymus and parathyroid development**.
- ❑ **Clinical manifestations:**
 - ✓ **Hypoplasia of the thymus** leading to **severe drop in peripheral blood T lymphocytes** → T cell deficiency (**normal B cell concentration in blood**)
 - ✓ **Abnormal facial features** (Long/narrow face and small mouth) in addition to cleft palate.
 - ✓ **Cardiovascular abnormalities**.
 - ✓ **Hypocalcemia** → Tetany (**low levels serum calcium is used for diagnosis**)



2. X-linked SCID:

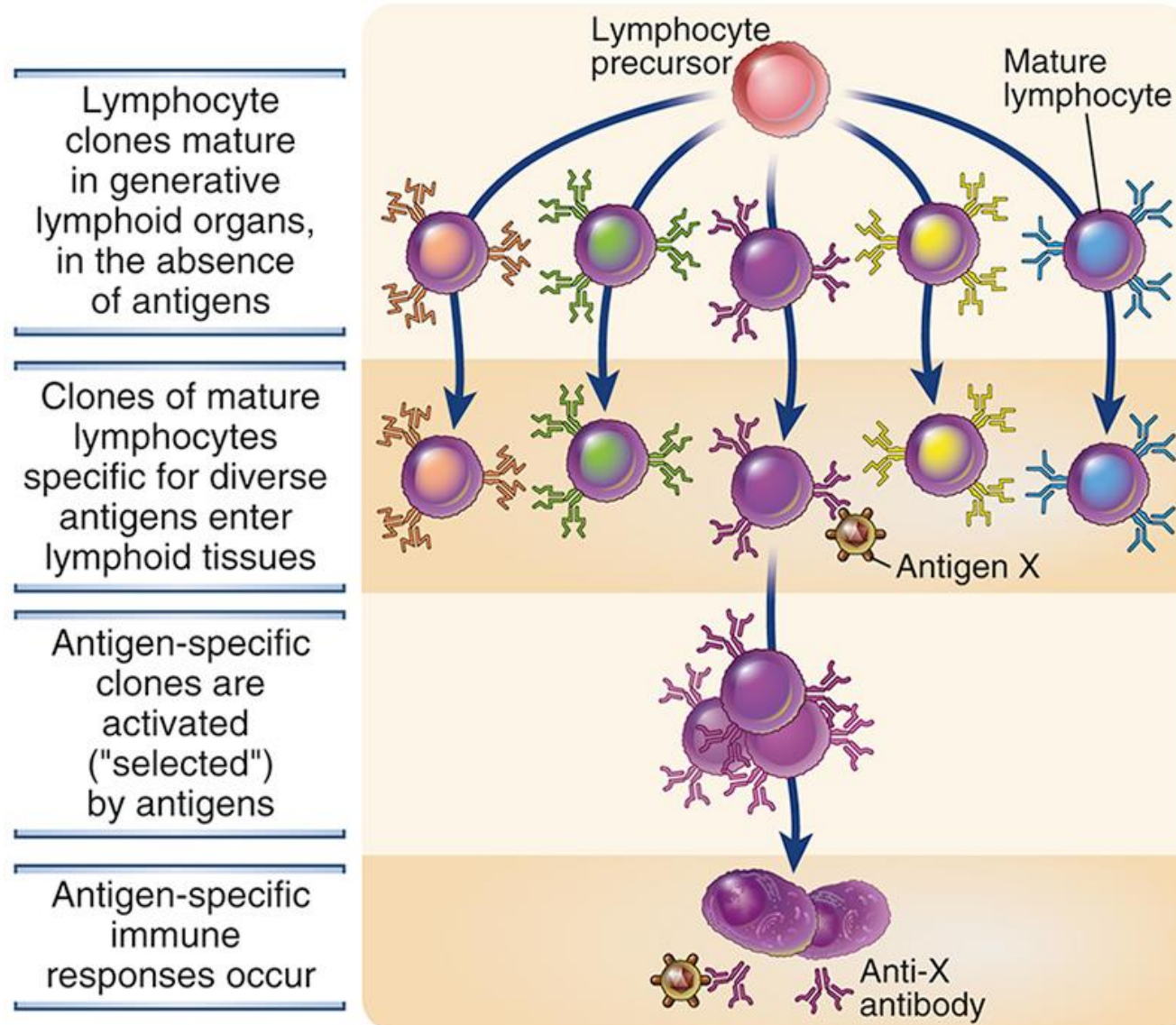
- ❑ **Genetic defect:** is caused by mutations in the gene encoding the common γ (γ_c) chain shared by the receptors for the cytokines IL-2, IL-4, IL-7, IL-9, and IL-15.
- ❑ **Immunological manifestations:** X-linked SCID is characterized by impaired maturation of T cells (IL-7 is required for T cell maturation) and NK cells (IL-15 is required for NK cell maturation) with a great reduction in their numbers, but the number of B cells is usually normal.



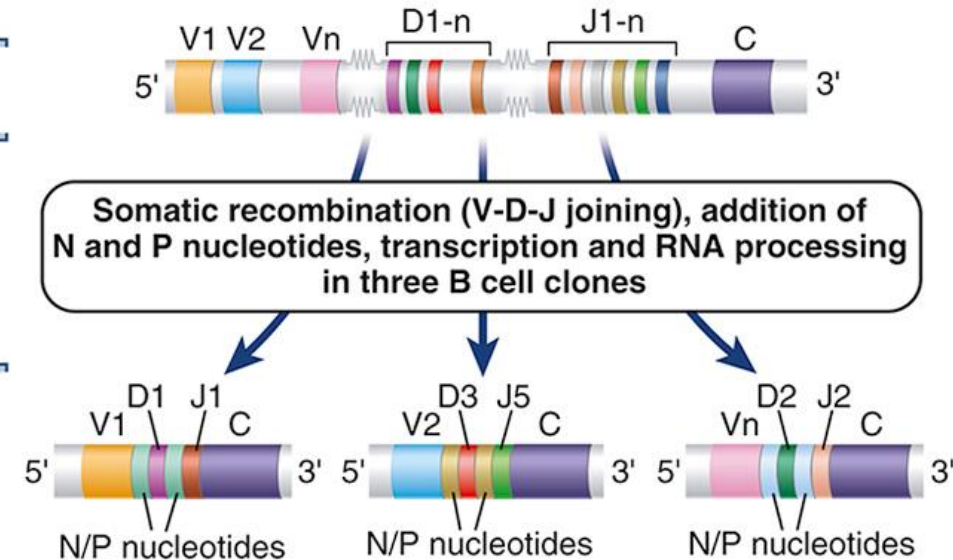
Source: Primer to the Immune Response. Ch. 15

Note: In X-linked disorders, the heterozygous females are usually phenotypically normal carriers, whereas males who inherit the abnormal X chromosome manifest the disease.

V(D)J recombination and the generation of diversity



Germline DNA



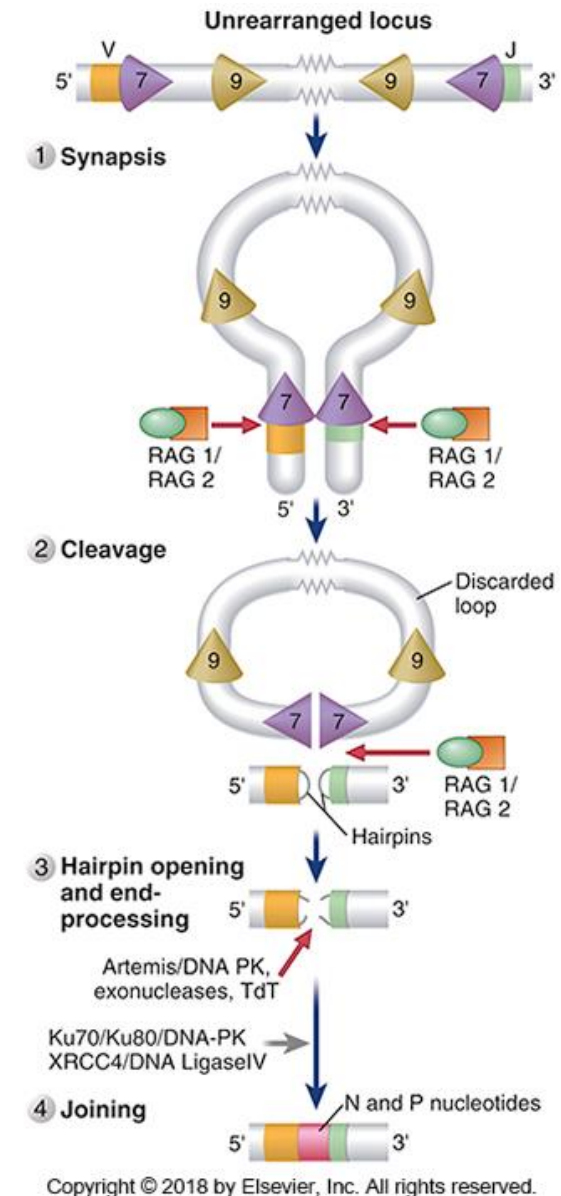
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3. SCID caused by defects in V(D)J recombination:

❑ Genetic defect:

- ✓ Mutations in the **RAG1** or **RAG2** genes that mediate the cleavage step during V(D)J recombination (**RAG SCID**)
- ✓ Mutation in **ARTEMIS** gene which encodes an endonuclease that resolves coding-end hairpins during V(D)J recombination (**Artemis SCID**)
- ✓ These mutations lead to the **absence of V(D)J recombination** process → **failure of expression** of the **pre-T cell receptor** and the **pre-B cell receptor**.

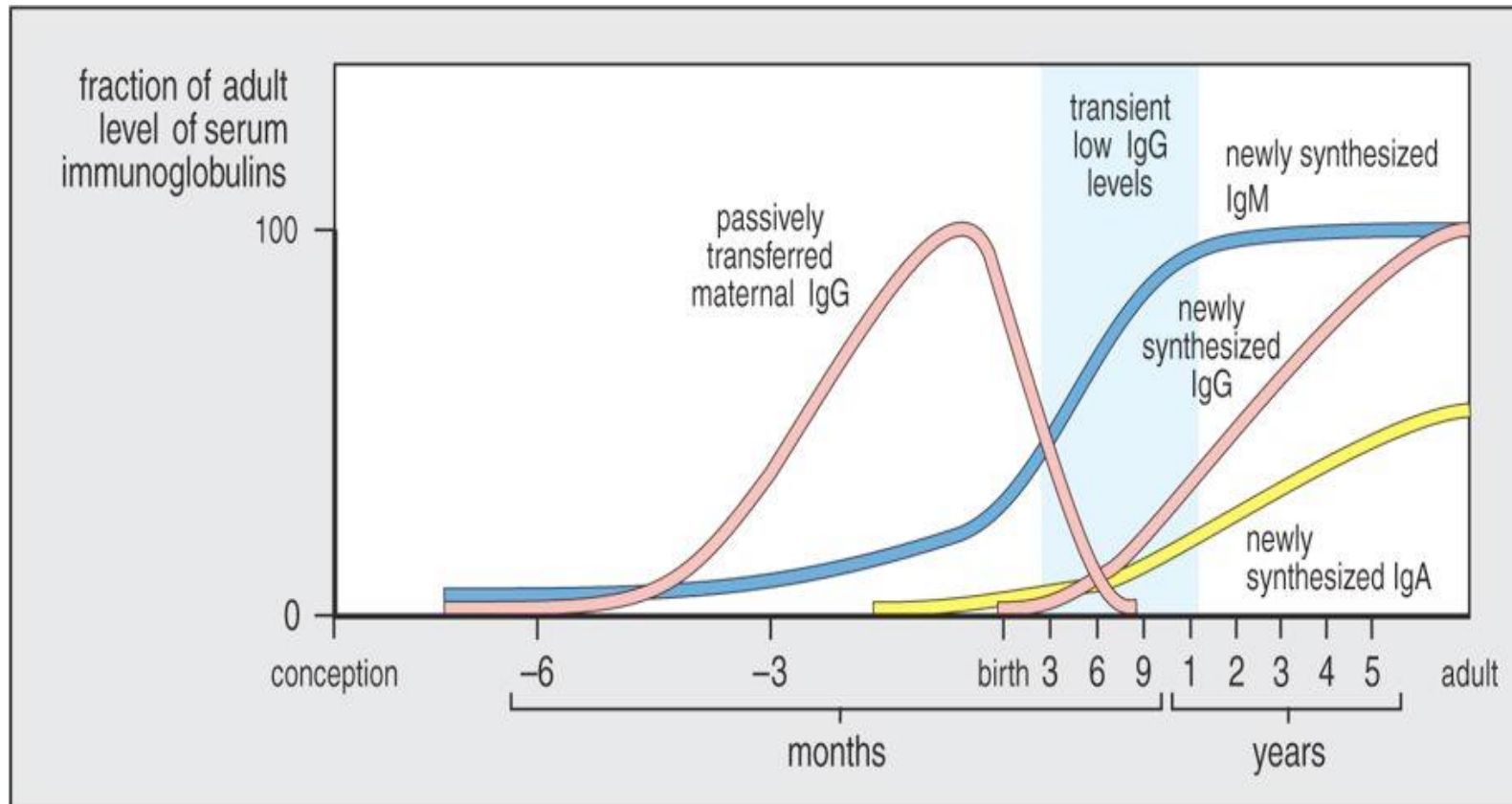
- #### ❑ Immunological manifestations: a **block in T and B cell development** causing SCID. In individuals with these mutations, B and T lymphocytes are absent and immunity is severely compromised.



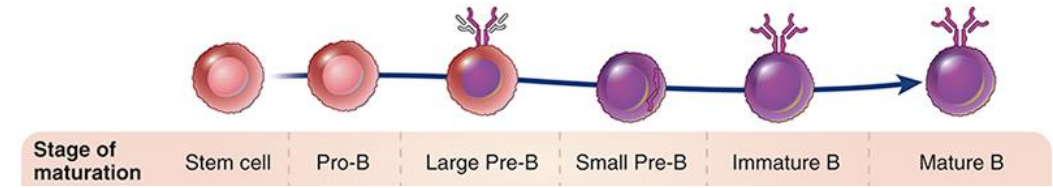
Clinical Cases of B Cell Immunodeficiency

Antibody deficiencies is often asymptomatic until 3-9 months due to maternal IgG that crosses the placenta and passively immunizes the embryo

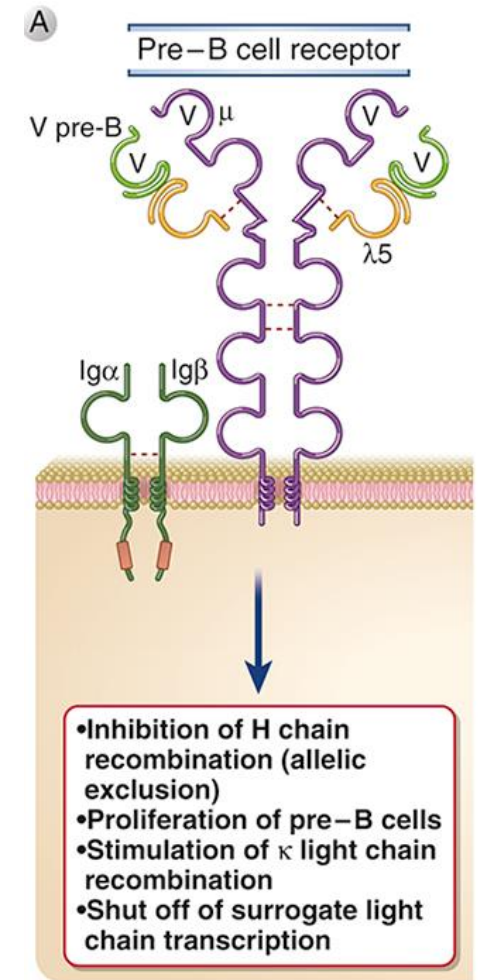
Note: Prematurely born babies will be born with lower IgG levels.



1. X-linked agammaglobulinemia (XLA) (Bruton agammaglobulinemia)



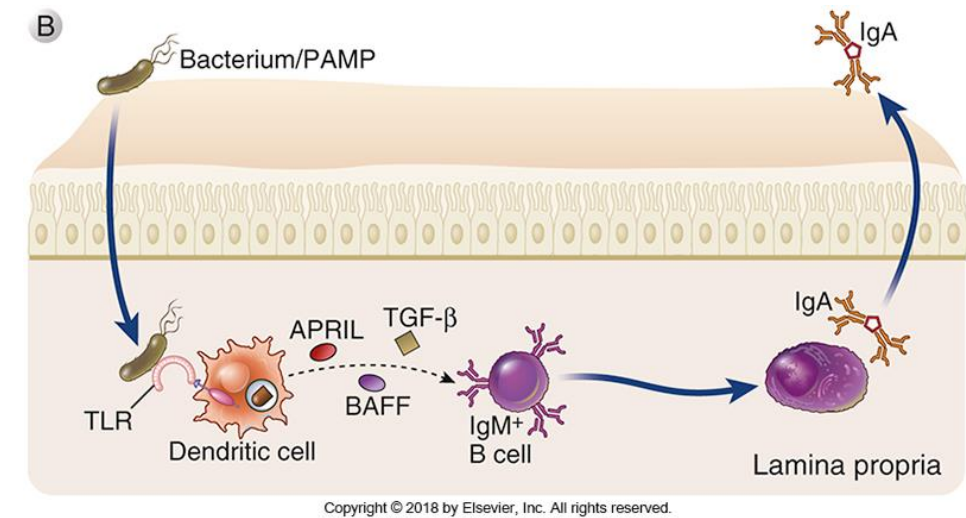
- ❑ **Genetic defect:** is caused by mutations or deletions in the gene encoding an enzyme called **Bruton tyrosine kinase (BTK)** → **failure of B cells to mature beyond the pre-B cell stage** in the bone marrow.
- ❑ **BTK is involved in transducing signals from the pre-BCR** that are required for the survival and differentiation of pre-B cells.
- ❑ **Clinical manifestations:**
 - ✓ **Low or undetectable antibodies.**
 - ✓ **Reduced or absent B cells** in peripheral blood and lymphoid tissues, **no plasma cells** and **germinal centers** in lymph nodes.
 - ✓ Tonsils are either absent or scant.
 - ✓ Susceptibility to **pyogenic encapsulated bacteria** such as ***Haemophilus influenzae*** and ***Streptococcus pneumoniae***.
 - ✓ Patients show recurrent **otitis media**, lung infections causing **pneumonia** and meningitis.
 - ✓ GI infections are also common.



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2. Selective IgA deficiency

- ❑ Is **the most common primary immunodeficiency** characterized by low serum IgA levels (<7 mg/dL).
- ❑ **Genetic defect:** Due to mutations of **TACI**, a receptor of **BAFF** and **APRIL** cytokines which stimulate T-independent class switching at mucosal tissues.
- ❑ **Clinical manifestations:**
 - ✓ **Many patients are asymptomatic**; (due to compensation by other antibody isotypes).
 - ✓ Some cases show severe, **recurrent respiratory tract** (Sinopulmonary) **or gastrointestinal infections, allergy, cancer** and **autoimmunity**.



Source: Cellular and Molecular Immunology, 9th edition. Ch. 14

Source: Parham Fig. 10.24

	Percentage of B cells making antibody of four different isotypes							
	Normal individuals				IgA-deficient individuals			
	IgA	IgM	IgG	IgD	IgA	IgM	IgG	IgD
Nasal glands	69	6	17	8	0	20	46	34
Lachrymal and parotid glands	82	6	5	7	0	21	22	57
Gastric mucosa	76	11	13	0	0	64	35	1
Small intestine	79	18	3	0	0	75	24	1

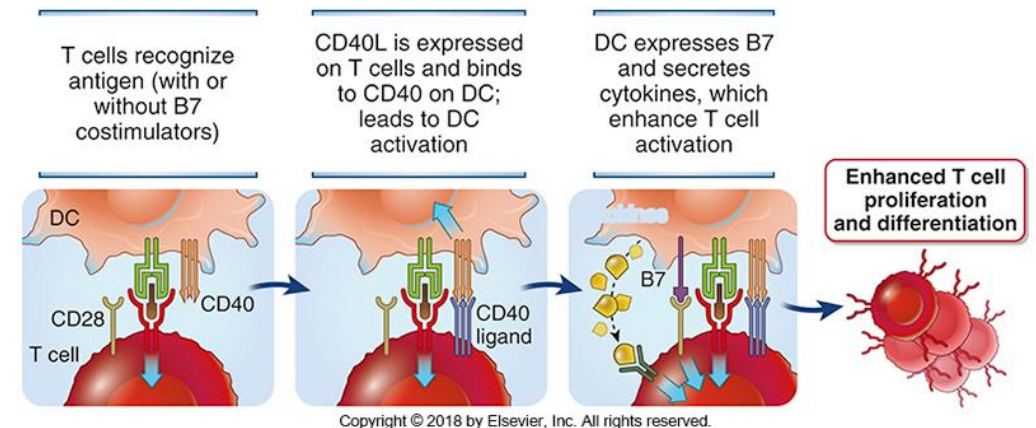
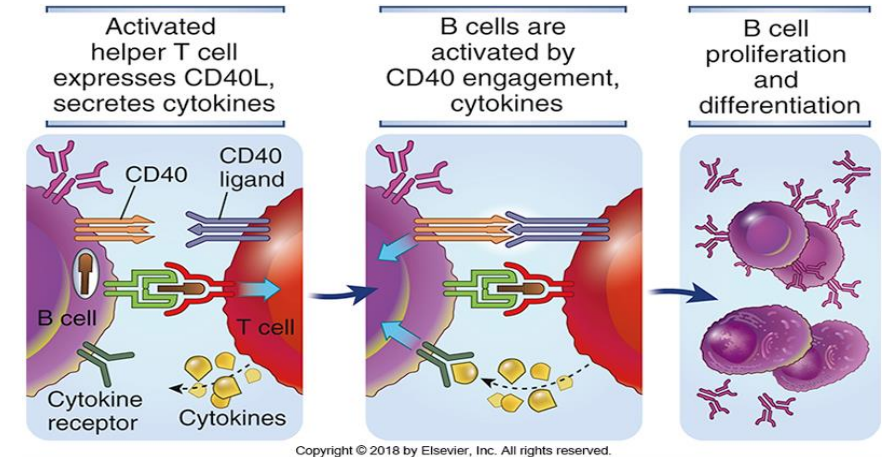
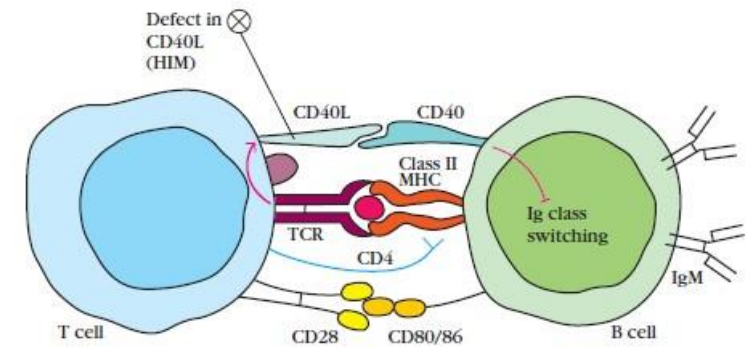
Note:

- ✓ IgA is found in **saliva** in the form of **secretory IgA (sIgA)**, which is the predominant antibody isotype in mucosal secretions such as saliva, tears, and mucus.
- ✓ sIgA is important for oral health because it helps protect the oral cavity from infections and contributes to maintaining a healthy oral microbiota.

T Cell Immunodeficiency

The X-linked hyper-IgM (HIM) syndrome:

- ❑ **Genetic defect:** is caused by mutations in the gene encoding the T cell effector molecule CD40 ligand (CD40L) → a defect to transduce signals through CD40 receptor → failure of heavy chain isotype switching, somatic hypermutation, germinal center formation and memory B cells development.
- ❑ **Clinical manifestations:**
 - ✓ Defective switching of B cells to the IgG and IgA isotypes, so the major isotype detected in the blood is IgM → increased susceptibility to infection by the extracellular bacteria, fungi and parasites.
 - ✓ Patients with HIM also show defects in cell-mediated immunity because CD40L and CD40 are also involved in T cell-dependent activation of macrophages and DCs.



Therapeutic approaches for congenital immunodeficiency

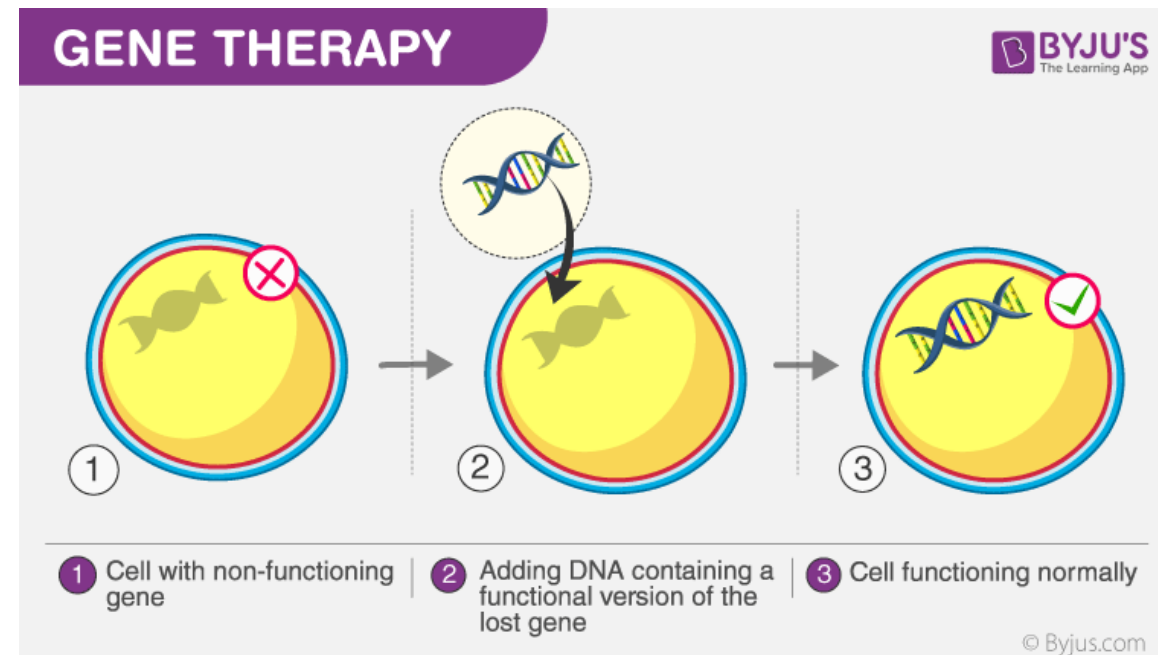
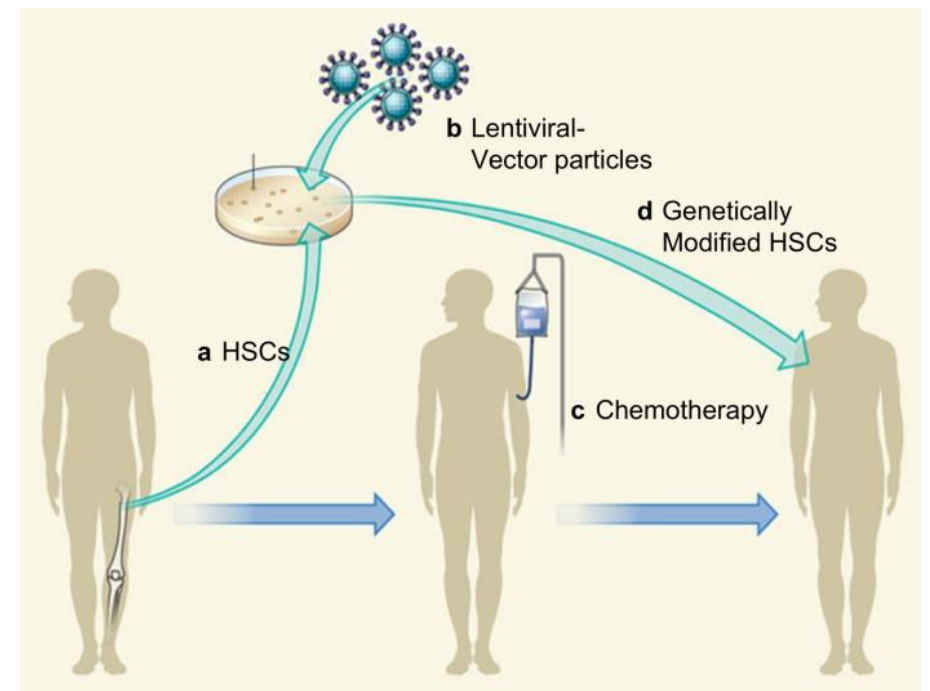
❑ Treatments for immunodeficiencies **have two aims:**

1. to minimize and control infections by treatment with Antiviral, antibacterial or antifungal.

2. to replace the defective or absent components of the immune system. **This approach includes;**

- **Passive immunization:** with pooled antibodies is very beneficial for agammaglobulinemic patients.
- **Hematopoietic stem cell transplantation** is currently the treatment of choice for many immunodeficiency diseases and has been successful in the treatment of SCID. It is most successful with careful T cell depletion and HLA matching to prevent graft-versus-host disease.
- **Enzyme replacement therapy:** Injection of the defective enzymes associated with immunodeficiency (E.g., RAG or Artemis SCID).

- **Gene Therapy:** ideal therapy for congenital disorders of lymphocytes is to **re-express the defective gene in self renewing stem cells**. This approach can be done by **isolating the patient's own HSCs** in the lab and **a normal copy of the gene is introduced** in these cells via a **viral vector** capable of **integrating** itself in **the host cell genome**.



Secondary (Acquired) Immunodeficiency

Mechanisms of secondary immunodeficiencies

1. Immunosuppression that occur as a biologic **complication of another disease process** e.g., **Malnutrition** and **Cancer**.
2. Immunodeficiencies may develop as complications of therapy for other diseases. Some **drugs such as corticosteroids** are given intentionally to **immunosuppress patients**, either for the **treatment of inflammatory diseases** or to **prevent rejection of organ allografts**.
3. Asplenia (loss of spleen).
4. Immunodeficiency may result from an **infection that target cells of the immune system**. E.g., **HIV**

TABLE 21.6 Secondary (Acquired) Immunodeficiencies

Cause	Mechanism
HIV infection	Depletion of CD4 ⁺ T cells
Protein-calorie malnutrition	Metabolic derangements inhibit lymphocyte maturation and function
Irradiation and chemotherapy for cancer	Decreased bone marrow lymphocyte precursors
Cancer metastases and leukemia involving bone marrow	Reduced site of leukocyte development
Immunosuppression for transplants, autoimmune diseases	Reduced lymphocyte activation, cytokine blockade, impaired leukocyte trafficking
Loss of the spleen due to trauma, sickle cell disease, or surgery	Decreased phagocytosis of blood-borne bacteria

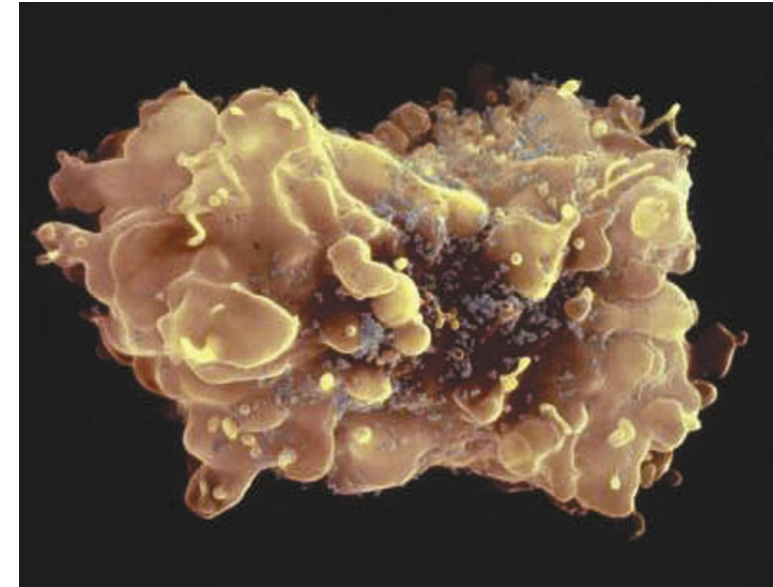
Source: Cellular and Molecular Immunology. 9th edition. **Ch. 21**

Human Immunodeficiency Virus (HIV) and the acquired immunodeficiency syndrome (AIDS)

- ❑ **AIDS** is the disease caused by infection with Human Immunodeficiency Virus (HIV) and is characterized by profound immunosuppression with associated opportunistic infections and malignant tumors, wasting syndrome (cachexia), and central nervous system degeneration.
- ❑ HIV is an infectious pathogen that evades host defenses by destroying the immune system.
- ❑ **HIV is transmitted from one individual to another by three major routes:**
 1. Sexual contact is the most frequent mode of transmission
 2. Mother-to-child transmission
 3. Inoculation of a recipient with infected blood or blood products
- ❑ HIV infection is easily prevented if precautions are taken such as use of gloves, screened blood products, clean needles and syringes.

Human Immunodeficiency Virus (HIV) and the acquired immunodeficiency syndrome (AIDS)

- ❑ **HIV tropism:** HIV infects primarily CD4+ T cells (picture attached) of the immune system, but it can infect other cells including macrophages, and dendritic cells.
- ❑ HIV is known as retrovirus because it can transform its RNA genome to DNA genome.
- ❑ Two closely related types of HIV, designated HIV-1 and HIV-2 are cytopathic (kill the cells they infect) for CD4 T cells. Infected Macrophages and DCs resist the cytopathic effect of the virus but help virus dissemination.
- ❑ HIV-1 is by far the most common cause of AIDS; HIV-2, which differs in genomic structure and antigenicity, causes a form of AIDS with slower progression than HIV-1–linked disease.



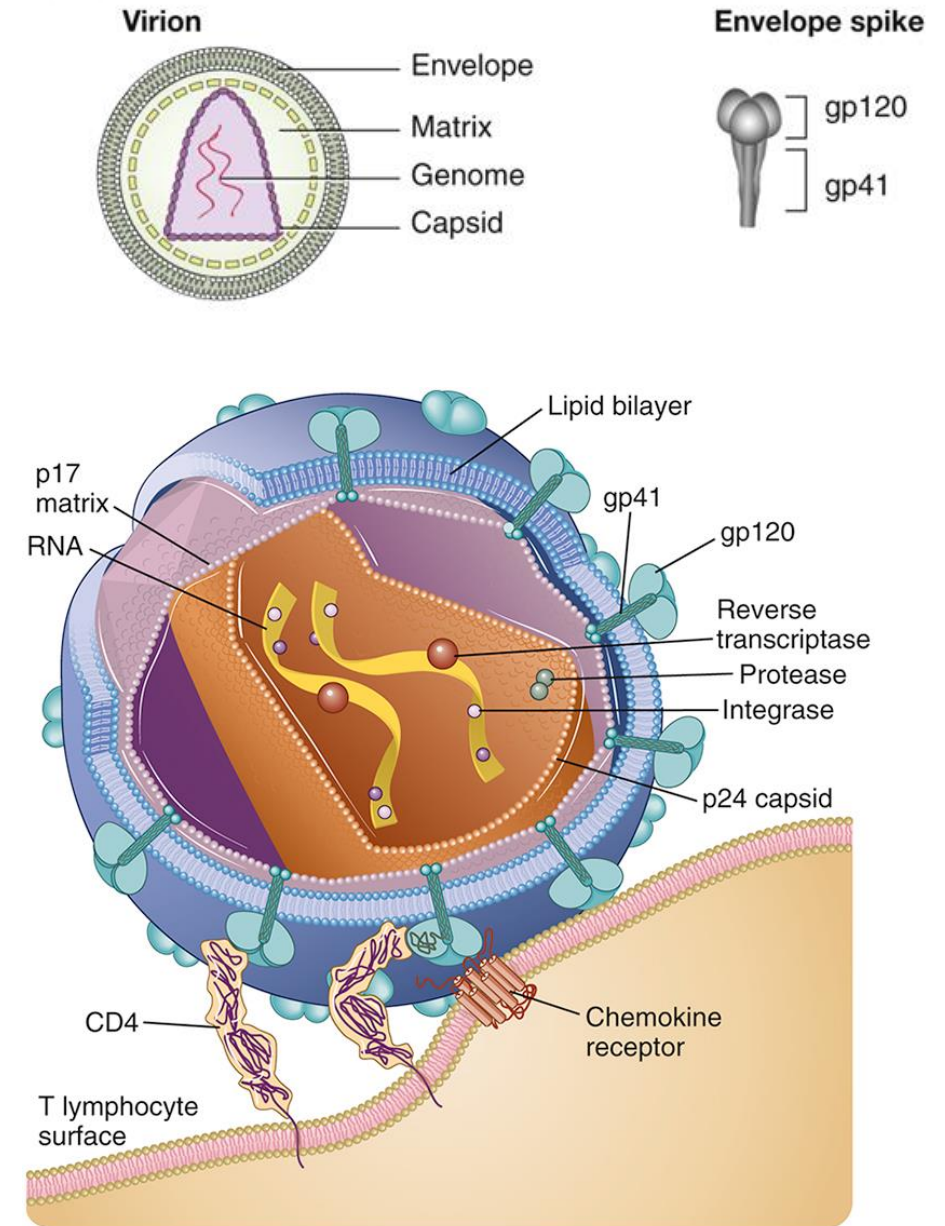
❑ HIV virion is composed of three layers;

1- **Envelope:** Outer **phospholipid bilayer**. The most prominent viral protein embedded in the envelope is the “**spike**” protein which is **made up of gp120 and gp41 glycoproteins**.

2- **Matrix:** The matrix is **composed of the viral structural protein p17** which acts as a scaffold structure around which the envelope is wrapped.

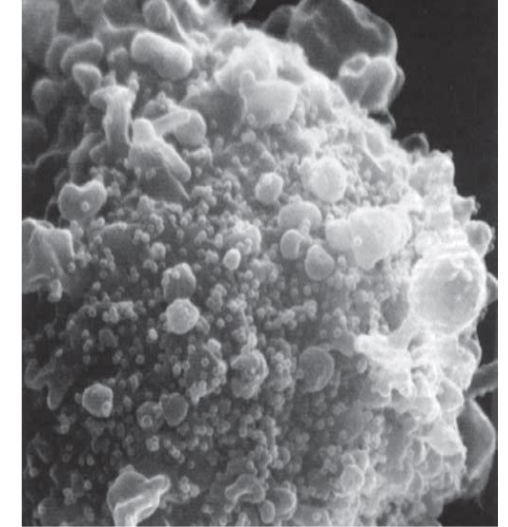
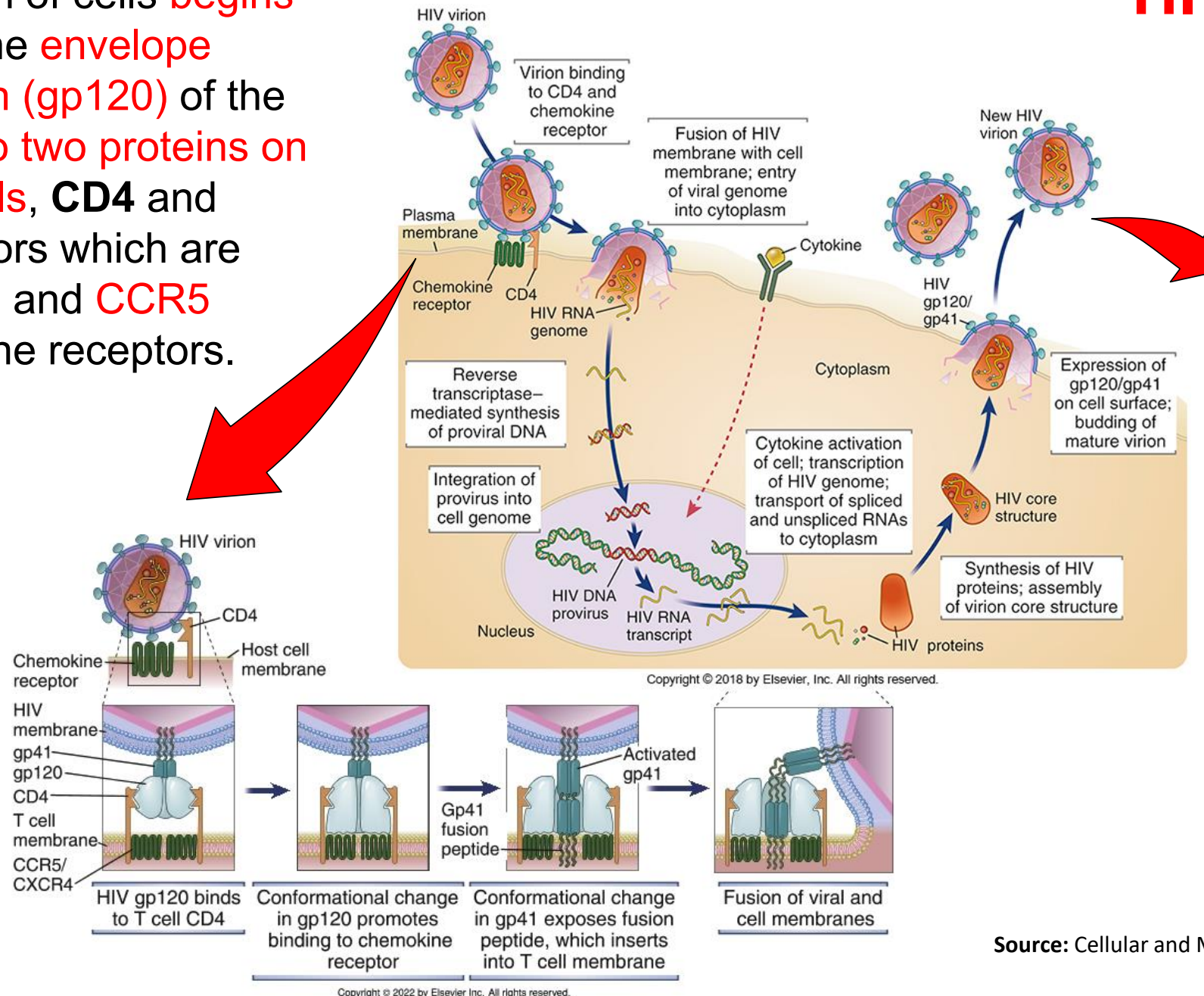
3- **Capsid:** The capsid is **the core of the virus** and made of the **structural protein p24** that **protects the two single strands of RNA** that make up the HIV genome.

Note: The **genomic viral RNA** within the capsid is associated with **viral reverse transcriptase (error prone)** which is required for provirus generation, **integrase** which is required for integration of the provirus in the host cell genome and **protease** which is involved in HIV assembly following reactivation.



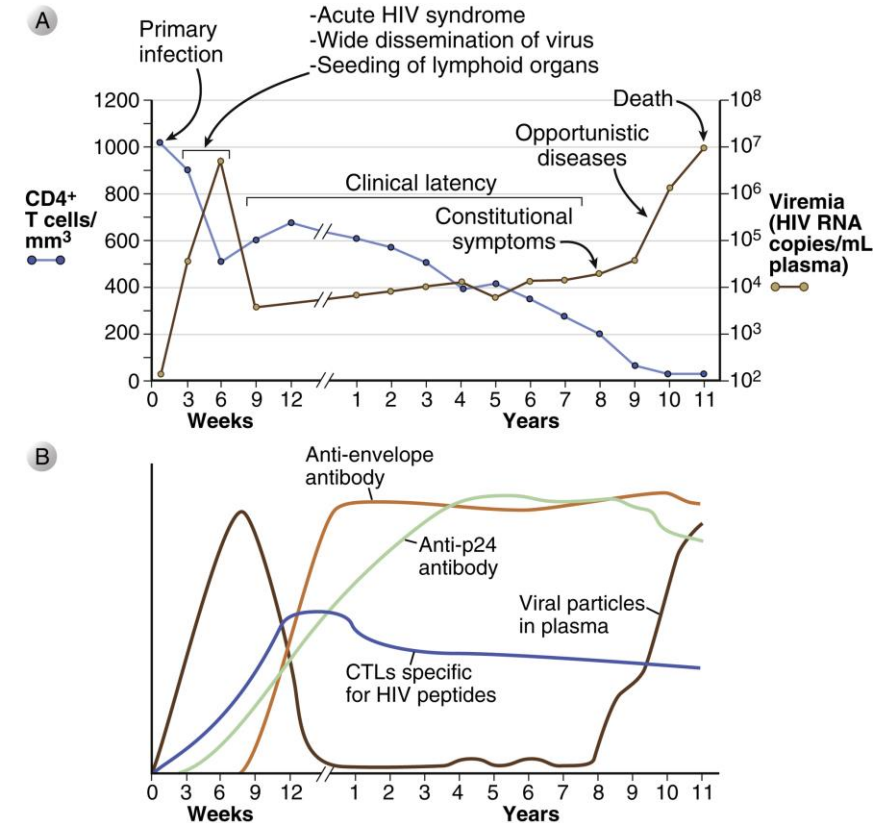
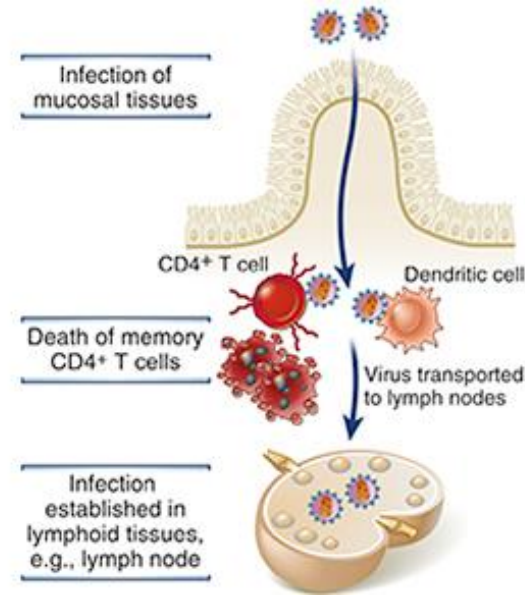
HIV life cycle

HIV infection of cells begins when the envelope glycoprotein (gp120) of the virus binds to two proteins on host cells, **CD4** and coreceptors which are **CXCR4** and **CCR5** chemokine receptors.



Immunopathogenesis of AIDS

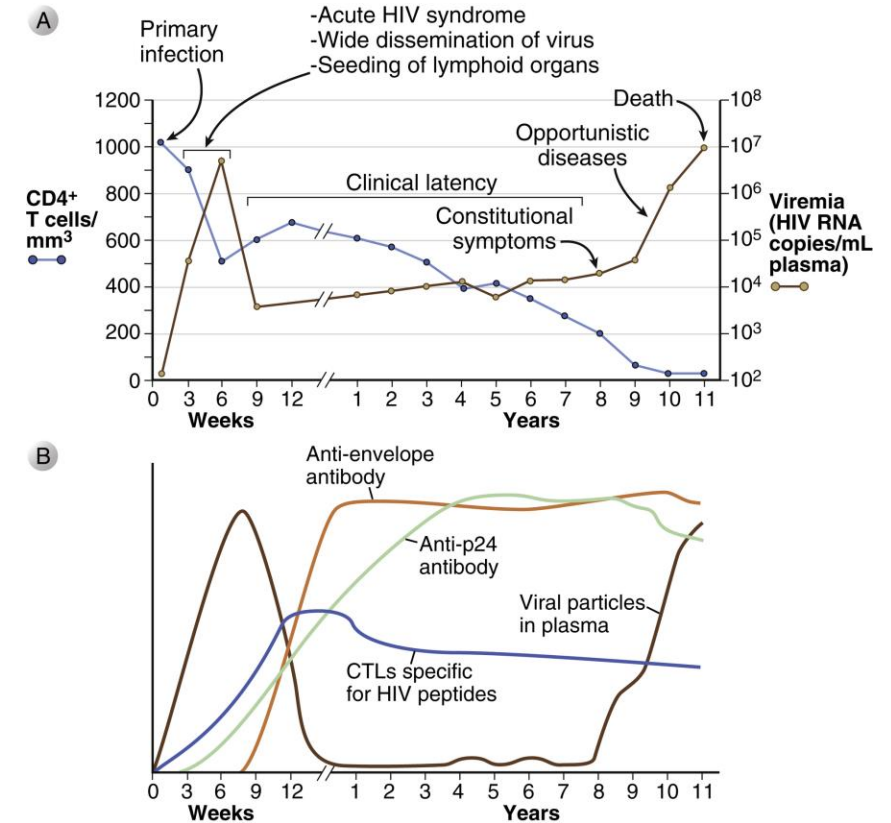
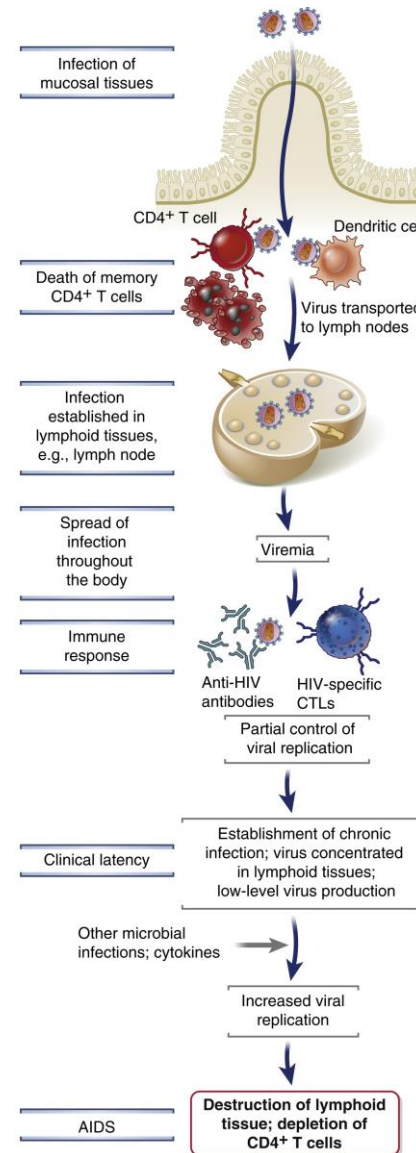
- ❑ HIV disease begins with acute infection, which is only partly controlled by the host immune response, and advances to chronic progressive infection of peripheral lymphoid tissues.
- ❑ Acute (early) infection is characterized by infection of CD4+ T cells, dendritic cells and macrophages in the mucosal tissues (sites of entry) and the death of many infected CD4+ T cells.
- ❑ The survived infected cells (dendritic cells and macrophages) carry the virus to the local lymph node and infect the resting CD4+ T cells.



Immunopathogenesis of AIDS (Cont.)

❑ The **transition** from the acute phase to the **chronic phase** of infection starts by **dissemination of the virus** from the local lymph node to the **whole body** and the **development of latency** where the viral genome (provirus) is integrated in the host genome and the viral replication slows down.

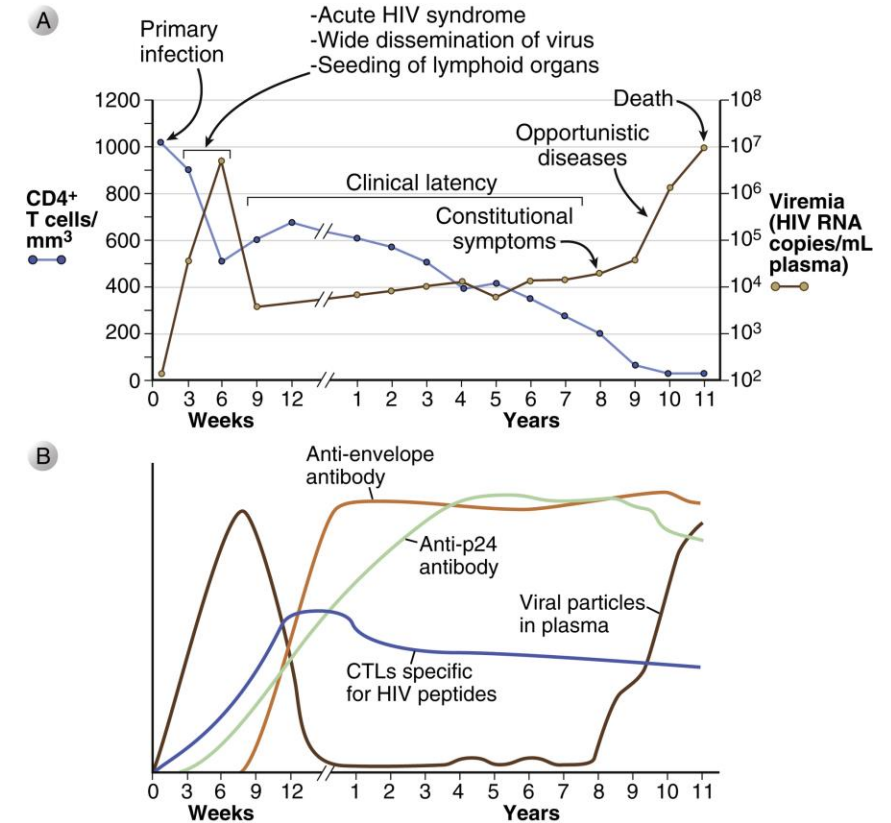
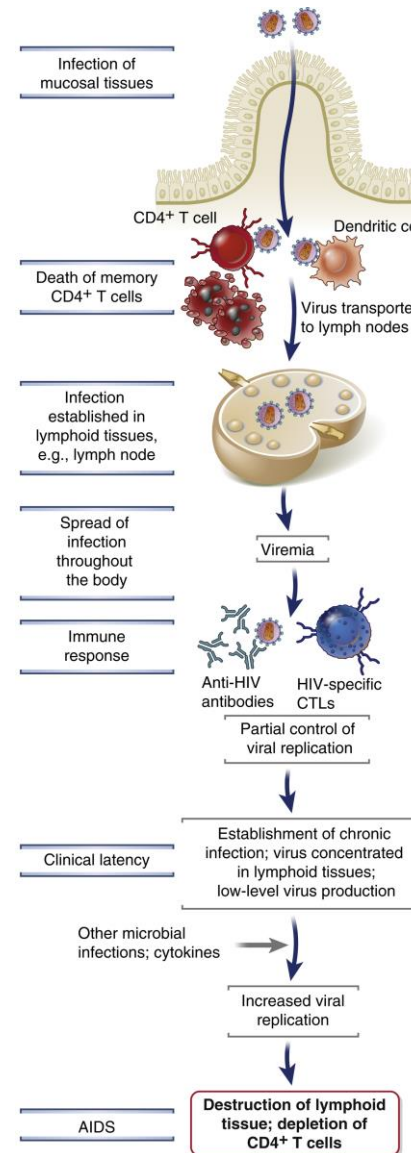
❑ During the **chronic phase** of the disease, **lymph nodes** and the **spleen** are **sites of continuous but slow HIV replication**.



Source: Cellular and Molecular Immunology. 9th edition. Ch. 21

Immunopathogenesis of AIDS (Cont.)

- ❑ Destruction of CD4⁺ T cells within lymphoid tissues steadily progresses during the latency period, and the number of circulating blood CD4⁺ T cells steadily declines.
- ❑ By the end of the latency period, the CD4⁺ T cells count (T4 count) falls below 200 cells/mm³ blood (25% of normal). At this late stage, the viral replication accelerates, and symptoms of AIDS appear



Source: Cellular and Molecular Immunology, 9th edition. Ch. 21

Mechanisms of Immunodeficiency caused by HIV

- ❑ HIV infection ultimately results in **impaired function** of both **the adaptive and innate immune systems**.
- ❑ The most prominent defects are in cell-mediated immunity, which results from the **destruction of CD4+ T cells**.
- ❑ There are **three major mechanisms** that contribute to the **loss of infected CD4+ T cells**:
 1. **Cytopathic effects** of viral infection.
 2. Killing by **antigen-specific cytotoxic T cells**.
 3. Activation of the **inflammasome** and elimination of infected cells by pyroptosis.
- ❑ **Macrophages and dendritic cells** are infected by HIV, **but they resist its cytopathic effect**. However, the ability of these cells to carryout **phagocytosis, antigen presentation and cytokine production is compromised by HIV infection** which contributes to the progression of immunodeficiency.

AIDS treatment

Anti Retroviral Therapy (ART)

HIV vaccine?

Currently, there is **no vaccine or permanent cure for AIDS** (due to antigenic variation, latency and infecting CD4 T cells), but quite effective **antiretroviral drugs** have been developed that **can control the infection**.

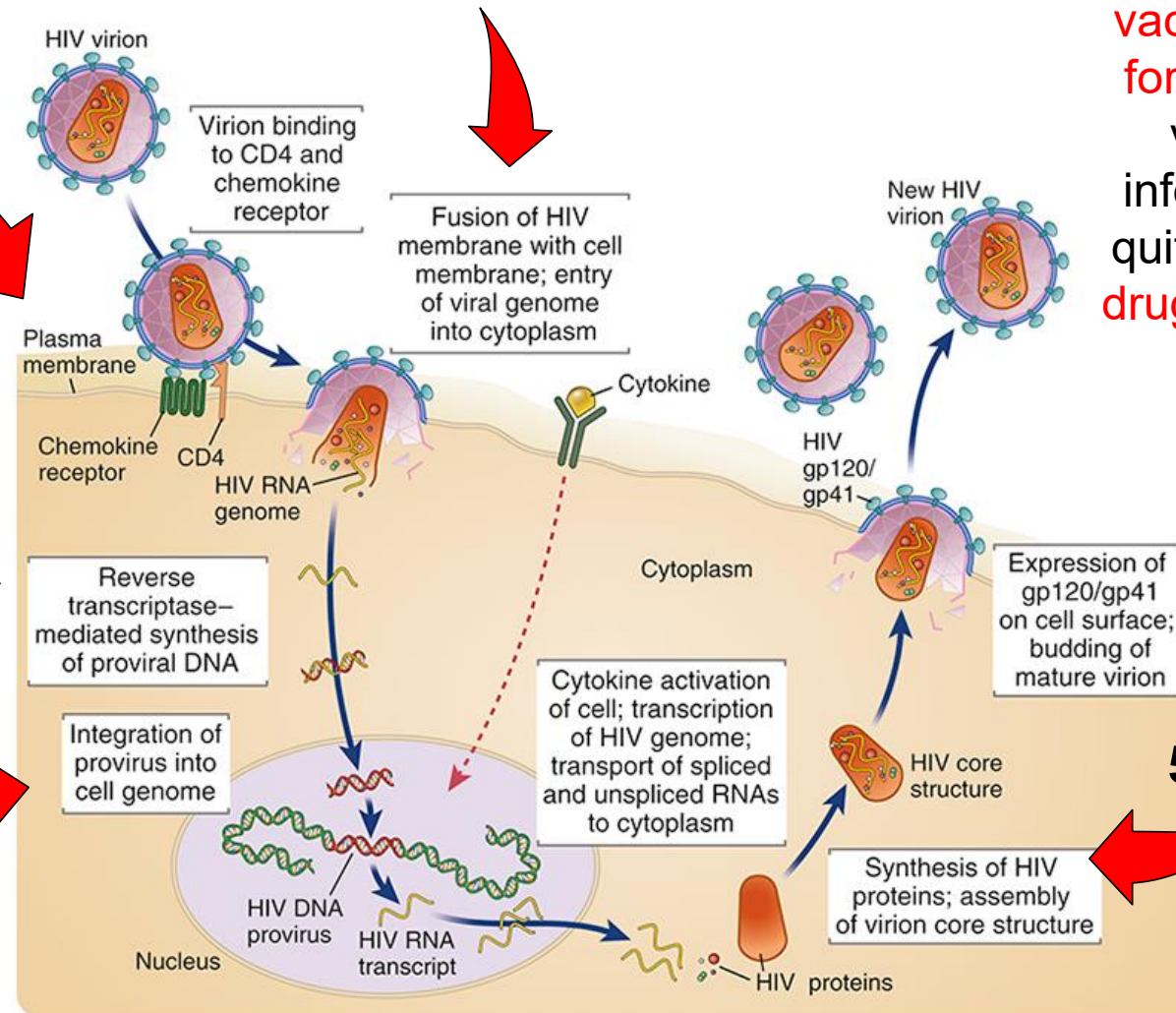
1. Chemokine receptor inhibitor (CCR5 blocker)

2. Fusion inhibitors.

3. Reverse Transcriptase Inhibitors

4. Integrase Strand Transfer Inhibitors (ISTIs)

5. Protease inhibitors



Take home message

- ❑ Immunodeficiency diseases are caused by congenital or acquired defects in adaptive and innate immunity. These diseases are associated with an increased susceptibility to infection and cancer.
- ❑ Severe combined immunodeficiencies (SCID) include defects in lymphocyte development that affect both T and B cells.
- ❑ Treatment of congenital immunodeficiencies involves transfusions of antibodies, stem cell transplantation, or enzyme replacement. Gene therapy may offer improved treatments in the future.
- ❑ Acquired immunodeficiencies are caused by infections, malnutrition, cancer, and immunosuppressive therapy for transplant rejection or autoimmune diseases.
- ❑ Several reservoirs of HIV exist in infected individuals, including CD4+ T cells, macrophages, and dendritic cells.
- ❑ HIV-induced depletion of CD4+ T cells results in increased susceptibility to infection by opportunistic microorganisms. In addition, HIV infected patients have an increased incidence of tumors
- ❑ AIDS can be treated by many drugs known collectively as Anti Retroviral Therapy (ART).